Niccolò Covoni

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Research Profile

Philosophy of Information, Philosophy of Physics, Metaphysics of Science, and Philosophy of Science.

Academic Position

2027 - 2030	**Fixed-Term Assistant Professor**, Sophia University Institute	*Loppiano, Italy*

Education

PhD candidate in Philosophy and in Research Methods in Science and Technology, Università della Svizzera italiana (USI) and University of Urbino Carlo Bo, 2023 - present. Dissertation: *The Reality of Entropy: A Foundational Study of Entropy and Information*. Supervisors: Professor Claudio Calosi and Professor Vincenzo Fano. **Expected thesis defense: February 2027**.

MS in Logic, Philosophy and History of Science, University of Florence, 2023, 110/110 cum laude. Dissertation: *Methods of Abstraction in Information Theory: The Case of Quantum-Like Machine Learning*.

BS in Philosophy, University of Pisa, 2020. Dissertation on the foundations of Birkhoff-von Neumann quantum logic.

Research Visits

07/26 - 08/26	**FTG Visiting Fellow**, Foundations of Thermodynamics Group, Nanyang Technological University; hosted by Professor Eugene Y. S. Chua	*Singapore*
02/26 - 04/26	**Visiting Student**, University of Lodz; supervised by Professor Hanna Podsędkowska	*Lodz, Poland*
01/25 - 03/25	**Visiting Student**, Rotman Institute of Philosophy, Western University; QISS project, supervised by Professor Francesca Vidotto	*Canada*
2024	**Visiting Student**, University of Cagliari; supervised by Professors Roberto Giuntini and Giuseppe Sergioli	*Cagliari, Italy*

Prizes & Grants

2025	**QISS Consortium Scholarship**, Western University	*Canada*
2023 - present	**PhD Scholarship**, Department of Pure and Applied Sciences, University of Urbino Carlo Bo	*Italy*
2020 - 2022	**Excellence Scholarship**, Percorso di Eccellenza Trasversale, University of Florence	*Italy*

Selected Publications

Book Under Contract

Niccolò Covoni, Andrea Di Biagio, Carlo Rovelli. 2026. *Tractatus Quanticus*. John Wiley & Sons Ltd. Under contract.

Articles, Chapters, and Proceedings

Niccolò Covoni, Marco Sanchioni. 2026. Relationality without Agents: Quantum Correlations and the Limits of QBism. In Anguel S. Stefanov and Vera Lyubenova (eds.), *Niels Bohr and Contemporary Philosophy of Science*, Minkowski Institute Press, pp. 61-86.

Vincenzo Fano, Niccolò Covoni. 2026. Phenomenological Elements in Carnap's *Aufbau*: The Epistemological Role of Sensible Intuition. *Sophia*, XVIII (2026-1), 29-51.

Niccolò Covoni, Marco Sanchioni. 2026. Relational Information Towards a New Kind of Information in Quantum Mechanics. *Foundations of Physics*, 56(1): 14.

Niccolò Covoni, Vincenzo Fano, Marco Sanchioni. 2026. Relational Quantum Mechanics and (Soft) Perspectivism. *Foundations of Physics*, 56(1): 6.

Niccolò Covoni, Flavia Marcacci, Gino Tarozzi. 2025. From Neo-Idealistic Historicism to Geymonat's Materialistic and Agazzi's Objectivistic Realism: The Roots of the Italian Philosophy of Science. *Bioethics Update*, 11(2), 129-140.

Niccolò Covoni. 2025. From Shannon to von Neumann: A Partial Understanding of Information. *Atti del XLIV Congresso Nazionale SISFA - Firenze 2024*, 89-94.

Niccolò Covoni, Alberto Corti, Vincenzo Fano. 2024. A Model-Based Form of Naturalised Metaphysics. *Revista Portuguesa de Filosofia*, 80(4), 783-818.

Niccolò Covoni, Gino Tarozzi. 2024. Empiricism and Realism Reconciled in Agazzi's Conception of Scientific Objectivity. *Distinctio: Journal of Intersubjective Studies*, 3(1), 47-55.

Niccolò Covoni, Giovanni Galli. 2025. A (Quasi-)Mathematical Theory of Public Language. *Proceedings of the 46th International Wittgenstein Symposium*.

Niccolò Covoni, Giovanni Macchia, Davide Pietrini, Gino Tarozzi. 2025. Non-Standard Realistic Models of Quantum Phenomena and New Forms of Complementarity. *Comptes Rendus de l'Académie Internationale de Philosophie des Sciences*, 3, 39-54.

Niccolò Covoni, Silvano Zipoli Caiani. 2024. Levels of Abstraction as Family Resemblances from the Classical to the Quantum Mechanical Representation of Reality. *Proceedings of the 45th International Wittgenstein Symposium*.

Selected Work in Review

Niccolò Covoni, Hanna Podsędkowska, Rafał Wieczorek. 2026. Segal Entropy and Relational Information.

Niccolò Covoni, Marco Sanchioni. 2026. Entanglement, Relative Facts, and the Meaning of Information in Relational Quantum Mechanics.

Niccolò Covoni, Marco Sanchioni. 2026. From "It from Bit" to Relational Quantum Mechanics: Wheeler, Rovelli, and the Relational Turn in Quantum Foundations.

Selected Presentations

Invited Talks

23/04/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics**. Universidad Complutense de Madrid, Spain.

19/04/2026. **Entanglement, fatti relativi e il significato di "informazione" nell'interpretazione relazionale della meccanica quantistica**. University of Urbino, Cesena, Italy.

11/03/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics**. Jagiellonian University, Krakow, Poland.

05/03/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics**. University of Lodz, Poland.

17/12/2025. **Once Again on von Neumann Entropy: Is It Always the Case That von Neumann Entropy Corresponds to Thermodynamic Entropy?**. Pontificia Universidad Católica de Valparaíso, Chile.

20/11/2025. **Relational Information: Towards a New Type of Information in Quantum Mechanics**. Institute for Quantum Optics and Quantum Information, Vienna, Austria.

Contributed Presentations

22/07/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics**. Asian Philosophy of Science Association Conference, NTU Singapore.

29/06/2026. **Entanglement, Relative Facts, and the Meaning of Information in Relational Quantum Mechanics**. SILFS Triennial International Conference, IUSS Pavia, Italy.

09/05/2026. **Segal Entropy and the Notion of Information**. Perspectives on the Foundations of Information Theory and Statistical Mechanics, USI, Lugano.

10/12/2025. **From "It from Bit" to Relational Quantum Mechanics: Wheeler and Rovelli in Historical and Conceptual Perspective**. Towards a Quantum Culture, University of Milan, Italy.

16/10/2025. **Relational Quantum Mechanics and (Soft) Perspectivism**. Philosophy of Science Conference for the 140th Anniversary of Niels Bohr's Birth, Sofia, Bulgaria.

Teaching

2026/2027	**Lugano Philosophy Colloquia**, Course for master's students	*USI, Lugano*
2026/2027	**Philosophy of Language**, Course for bachelor's students	*Sophia*
2026/2027	**Logic, Game Theory and Happiness**, Course for master's students; co-taught with Marco Sanchioni	*Sophia*
2026	**Introduction to the Philosophy of Thermodynamics**, Introductory course for philosophy students	*USI, Lugano*
2025/2026	**Philosophy of Language**, Teaching Assistant to Professor Marco Sanchioni	*Sophia*
2024	**Epistemology**, Teaching Assistant to Professor Gino Tarozzi	*Urbino*
2024	**Introduction to Edutainment**, Lecture for master's students in Communication Science	*Florence*
2023	**Introduction to Description Logic**, Lecture for master's students in Philosophy	*Urbino*

Academic Service

Coordinator, CI^2 - Complexity & Information Initiative, ICTS, Sophia University Institute; with Marco Sanchioni as co-coordinator, 2026 - present.

Editor, Scientific Committee, *Sophia: Research on Fundamentals and Correlation of Knowledge*.

Referee for *Foundations of Science*, *Synthese*, *Isonomia*, *The Reasoner*, *Humana.mente*, *The European Physical Journal H*, *The British Journal for the Philosophy of Science*, *Humanities and Social Sciences Communications*, and *Philosophical Inquiries*.

Organiser, Perspectives on the Foundations of Information Theory and Statistical Mechanics, USI, Lugano, 2026.

Organiser, International Summer School in Philosophy of Physics, University of Urbino, since 2024.